

Conditional Generative Models

Generative models, for example GAN or VAE, can learn conditional distributions. Recall that GAN plays the min-max game:

$$\arg \min_{\theta} \max_{\phi} \mathbb{E}_{x,z} [\ln(1 - D_{\phi}(G_{\theta}(z))) + \ln D_{\phi}(x)] .$$

In the context of the MNIST dataset, there are labels Y associated with images X and there are two natural ways to form conditional probabilities between X and Y : $p(X|Y = y)$ or $p(Y|X = x)$. Note that $p(Y|X = x)$ is the distribution of the labels given $X = x$, which, in the case of MNIST, is deterministic, and is therefore not very interesting. The other case, $p(X|Y = y)$, is the distribution of X (the images) given a label y . If this conditional distribution is learned, we can generate hand-written digits according to a specific label.

To learn this conditional distribution, we modify generator $G_{\theta}(z)$ to include the information of the conditioned label y : $G_{\theta}(y, z)$. Our discriminator becomes $D_{\phi}(x, y)$. The min-max game becomes

$$\arg \min_{\theta} \max_{\phi} \mathbb{E}_{x,y,z} [\ln(1 - D_{\phi}(G_{\theta}(y, z), y)) + \ln D_{\phi}(x, y)] .$$

The conditional VAE follows similarly by conditioning on the labels y . The prior of the latent variable z remains the same $p_{\theta}(z)$. We update the likelihood to be $p_{\theta}(x|z, y)$. The posterior is therefore $q_{\phi}(z|x, y)$. We reparametrize by $z = g_{\phi}(x, y, \varepsilon)$ for $\varepsilon \sim p(\varepsilon)$.